Hall Ticket Number:

MR-22

Code: B425CS4S

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. IV Semester Supplementary Examinations, December 2025 / January 2026

Database Management Systems

(Common to CSE, IT, CSB, CSM and CSD)

Time: 3 Hours | ***Answer ALL Questions*** | Max. Marks: 60

Part-A (10 x 1 = 10 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	What is the Data Model?	1	1	1	1,2
1. b)	What is Database Design?	1	2	1	1,2
1. c)	What is Domain relational calculus?	1	2	2	1,2
1. d)	What is Relational Algebra?	1	2	2	1,2
1. e)	Describe about form of basic SQL query.	1	2	2	1,2
1. f)	Define triggers and active databases.	1	2	3	2,3
1. g)	Define implementation of Isolation.	1	4	4	1,2
1. h)	Define Log-Bases Recovery.	1	2	4	1,3
1. i)	Discuss about data on external storage.	1	2	5	2,3
1. j)	Discuss about cluster indexes.	1	3	5	1,2

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Explain Levels of Abstraction in a DBMS.	5	2	1	1,2
2. b)	With a neat diagram explain Structure of a DBMS.	5	2	1	1,2
	OR				
2. c)	Describe about entity, entity set, Relationship and Relationship Sets.	5	2	1	1,2
2. d)	Explain about additional features of ER Model.	5	2	1	1,2
3. a)	Describe Integrity constraint over relations.	5	2	2	2,3
3. b)	Explain enforcing integrity constraints in Data Base.	5	2	2	1,2
	OR				
3. c)	Explain about Logical database design, altering tables and views.	10	2	2	1,2
4. a)	What are Fourth normal form and Fifth normal form in Data bases?	5	3	3	1,2
4. b)	Analyse about BCNF, lossless join decomposition.	5	5	3	1,2
	OR				
4. c)	Explain about multivalued dependencies.	5	3	3	1,2
4. d)	Describe about Nested queries, aggregation operators and NULL values.	5	2	3	3,4
5. a)	Explain about Transaction concept and Transaction state.	5	2	4	1,2
5. b)	Describe about implementation of Atomicity and Durability.	5	3	4	1,2
	OR				
5. c)	Describe about Concurrent Execution and Serializability.	5	4	4	1,2
5. d)	Describe about testing for serializability and Lock based protocols.	5	4	4	3,2
6. a)	Analyse about Tree based Indexing and comparison of File Organizations.	5	3	5	2,3
6. b)	Explain about Index data structures and Hash based Indexing.	5	2	5	1,2
	OR				
6. c)	Explain about B+ Trees with respect to dynamic index structure.	5	5	5	1,2
6. d)	Discuss about Intuitions for tree indexes and ISAM.	5	3	5	3,2

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome